

# Mehfooj Alam

## AI Systems & Backend Engineer | Agent Runtimes • Async Python • Developer Tools

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### PROFESSIONAL SUMMARY

Software Engineer building **agent runtimes and developer tools from first principles**: ReAct execution loops, tool calling, deterministic orchestration, and **async Python backends** (FastAPI, PostgreSQL, Redis, Docker). Experience with CI/CD infrastructure, log analysis pipelines, and evaluation environments for LLM agents.

### TECHNICAL SKILLS

**Languages**: Python, TypeScript, JavaScript, SQL, Bash, (*Exposure: Go*)

**Backend & Systems**: FastAPI, Async Python, PostgreSQL, Redis, Alembic, Node.js, Express, REST APIs, CLI Tools

**AI/ML & Agent Systems**: Agent Runtimes, ReAct Loops, Tool Calling, Deterministic Orchestration, Evaluation Harnesses, Structured Outputs (Zod, Pydantic), RAG, LangChain, LangGraph, PyTorch

**MLOps & Infrastructure**: Docker, Docker Compose, Podman, GitHub Actions, CI/CD, MLflow, Great Expectations, Distributed Systems, Observability

### PROFESSIONAL EXPERIENCE

#### Backend Engineering Trainee

Jul 2026 – Present

FlyRank AI

Remote

- Ship async REST APIs ([repo](#)) in near daily commits with **FastAPI, PostgreSQL, and Redis**: connection pooling, Alembic migrations, request validation, search and filter endpoints, and a health check covering both datastores.
- Containerized the stack with **Docker Compose** (migrations run on startup) and wrote an integration test suite (13 passing tests on an isolated test database); fixed a duplicate insert race condition with database level constraints.
- Currently adding **JWT authentication** through an identity provider, protected routes, and **Redis backed rate limiting** middleware.

*Technologies Used: Python, FastAPI, PostgreSQL, Redis, Docker Compose*

#### Software Engineer Intern (Backend, Agent Systems & OSINT)

May 2026 – Jul 2026

AiGeeks

Remote

- Designed and built an initial lightweight **agentic orchestration layer** [*idea similar to ([openosint.tech](#))*] on native **Anthropic tool use APIs**: the model only selects from registered tools, while execution follows **deterministic rules** that remove hallucinated outputs and constrain tool invocation.
- Architected a 5 layer **OSINT reconnaissance pipeline** (passive discovery, active resolution, web surface probing, threat intelligence) delivering structured output through **Zod schema validation**.
- Built **low latency**, deduplicated ingestion microservices for username, phone, email, website, and domain targets on the [Intelfetch](#) platform.
- Led the architectural foundation and deployment of the end to end **CI/CD infrastructure** with Docker and GitHub Actions.

*Technologies Used: TypeScript, Python, Anthropic Tool Use, Node.js, PostgreSQL, Docker, GitHub Actions, CI/CD*

#### Data Analyst Intern

Mar 2025 – Jul 2025

KultureHire

Remote

- Built automated data-cleaning and ETL analysis pipelines; derived actionable performance metrics from large candidate datasets to improve product SLAs.

*Technologies Used: Python, SQL, PostgreSQL*

### PROJECTS

#### LLM Engine | Terminal Native Coding Agent Runtime (versioned releases, ongoing)

- Built** an **agent runtime** from first principles: a manual **ReAct execution loop**, a unified **tool registry**, and native provider **tool calling** with XML fallback, both normalized into one invocation format across Groq and OpenAI models.
- Exposed through a decoupled **FastAPI** gateway and a **terminal native** interface, with coding tools for file reading, directory listing, and shell execution.
- Shipping** in versioned phases (v0.4.0 complete); next: **autocommit CLI**, interactive coding agent, **evaluation harness**, and persistent memory.

#### LogIQ | Distributed CI/CD Log Analysis & Fault Diagnostics Engine

- Engineered** an async **FastAPI** service for automated **CI build failure diagnosis**, streaming logs through chunked buffers and background workers with **sub-300ms preprocessing latency** on large builds.
- Designed** tiered failure routing (regex heuristics, **sentence embeddings**, LLM reasoning) to cut unnecessary inference; containerized with **Docker**, with fine tuned classifiers in development.

#### RL Code Review Agent | OpenEnv Reinforcement Learning Environment

- Engineered** a 3 stage RL curriculum for autonomous Python code review covering **PEP8, security, and algorithmic complexity**, with reward driven task progression.
- Implemented** a sandboxed **evaluation harness** combining unit tests and **Ruff** analysis into reward signals; integrated Qwen inference with OpenEnv and Dockerized deployment.

#### Fullstack-ML | Pipeline Automation & Infrastructure Platform

- Architected** a reproducible ML pipeline with automated validation (**Great Expectations**) and experiment tracking (**MLflow**).
- Automated** CI/CD with **GitHub Actions**, Docker builds, and deployment of a FastAPI/XGBoost inference service with a Gradio front end.

### EDUCATION

#### Bachelor of Technology in Computer Science (Data Science)

2023 – 2027

St. Andrews Institute of Technology & Management, Gurgaon | MDU Rohtak